Computer Vision Algorithms for Human Activity Recognition

Special Subject
Maria Apostolidou

Supervisor :
Potamianos Gerasimos
University of Thessaly
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Abstract

This work investigates vision-based human activity recognition (HAR) for smart home scenarios, comparing appearance features from RGB video with motion information derived from skeleton keypoints on a curated Kinetics subset consisting of seven thousand clips and four hundred actions. Five CNN-based RGB models and three skeleton-based approaches, relying on keypoints from MediaPipe, HRNet, and PIXIE, were evaluated, with Top-1 and Top-5 accuracy reported on a separate test split. Overall, RGB approaches achieved higher accuracy than skeleton ones, with MobileNet V2 performing best among RGB models. Within skeleton methods, HRNet reached the highest Top-1 score, whereas MediaPipe achieved the strongest Top-5 results, highlighting complementary advantages between precise 2D landmarks and compact 3D landmarks with visibility scores. These findings indicate that lightweight spatial encoders with simple temporal aggregation can serve as solid baselines in limited data settings, while skeleton representations remain appealing for privacy-conscious smart home applications.

1 Introduction

The automatic identification of physical actions carried out by people using information gathered from sensors or visual inputs is known as human activity recognition (HAR). Given the continuous evolution of computer vision, HAR has become a fundamental task with numerous applications in interactive systems, smart environments, surveillance, and healthcare monitoring. It improves the contextual awareness and responsiveness of intelligent systems by empowering machines to interpret complex human behavior in real time [18, 3, 20].

Variations in human pose, appearance, background clutter, camera angles, and temporal dynamics make the HAR problem especially difficult. Early methods relied heavily on handcrafted representations, including optical flow, motion history images, and silhouette-derived descriptors. However, these methods often struggled to generalize across different action classes or environments. [18, 2].

The capabilities of HAR systems have been greatly enhanced by the development of deep learning, especially through Convolutional Neural Networks (CNNs) and attention-based models. In order to extract low-level and mid-level spatial features, including textures, edges, body outlines, and motion cues embedded in the pixel intensities, CNNs work directly on raw RGB frames. Although these features have a lot of visual content, they are frequently mixed up with variations in appearance and background. Skeletal-based methods, on the other hand, use pose estimation frameworks to extract high-level, semantically structured features that characterize the joint configurations and posture of humans in 2D, 3D, or 6D space. Skeletal representations demonstrate greater robustness to background noise, illumination changes, and occlusions by discarding visual appearance information and emphasizing only body geometry and motion. To assess their efficacy in human activity recognition, this project investigates two modalities: structured skeleton features derived from pose estimation and raw visual features learned via CNNs [18, 16, 2, 12].

This project investigates and compares two pipelines for human activity recognition using RGB videos:

1. The first approach involves extracting spatial features from RGB video frames using various CNN-based (e.g., ResNet, VGG, MobileNet), which are then used for activity classification [19, 4, 9, 16, 7].

2. The second approach leverages pose estimation techniques (MediaPipe, HRNet, PIXIE) to extract skeleton-based representations to analyze the sequence of joint features for recognizing activities [15, 14, 11].

The work's potential for smart home settings, where real-time comprehension of human activity can facilitate automated security monitoring, intelligent assistance, and fall detection, is a major driving force. The goal of this project is to find computationally efficient and successful HAR strategies that can be implemented in actual smart systems by comparing various architectures and data modalities [1, 13, 8, 20].

The rest of this report is structured as follows: Section 2 reviews related work, while Section 3 details the proposed methodology. Section 4 describes the experimental framework, and Section 5 reports the results. Section 6 discusses potential applications in smart home contexts, and finally, Section 7 concludes with a summary and directions for future research.

2 Related Work

2.1 Human Activity Recognition

The goal of the crucial and developing field of human activity recognition (HAR) in computer science and pattern recognition is to automatically recognize, categorize, and comprehend human actions, behaviors, and intentions from data. It is predicated on systems' capacity to comprehend and classify human behavior seen in still photos, videos, or signals from different sensors. Video surveillance systems, robotics and human-computer interaction (HCI) for characterizing human behavior, automated health monitoring, smart home development for assisted living (AAL), sports analysis, and anomalous activity detection are just a few of the many applications for HAR [18, 10, 12, 20, 4].

Human activities can be categorized based on their complexity into six levels: (i) gestures, (ii) individual actions, (iii) human-object or human-human interactions, (iv) group actions, (v) behaviors, and (vi) events .In this context, "activity" describes a sequence of actions corresponding to a specific body movement, while "behavior" is used to characterize both activities and events associated with gestures, emotional states, facial expressions, and auditory stimuli of an individual [18, 17, 9].

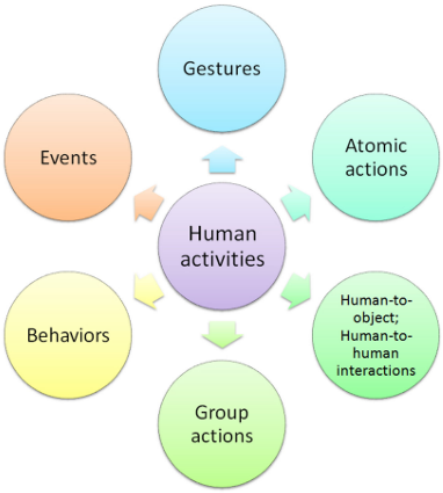

Figure 1: Decomposition of human activities [18].

HAR systems can be based on unimodal or multimodal approaches , depending on the nature of the sensor data they use. Data sources include camera sensors (such as RGB video, depth and infrared data), wearable sensors (such as inertial measurement units - IMUs, gyroscopes and accelerometers) and environmental sensors . A closely related field is Human Pose Estimation (HPE) , which aims to identify and locate key points on the human body to extract pose [5, 15, 3].

However, developing a fully automated and low-error HAR system is a complex and demanding task , due to issues such as background clutter, partial occlusion, changes in scale, perspective, lighting, appearance, frame resolution, and inherent intra- and inter-category similarities. Furthermore, the time-consuming and subjective nature of data annotation, the lack of adequate and realistic reference datasets, and the need for generalization and robustness to external factors, constitute significant challenges. Deep Learning (DL) has emerged as a powerful tool to address these limitations, allowing for the autonomous extraction and enhancement of features from raw data,

without the need for manual feature engineering. DL approaches can handle complex and dynamic activity patterns, significantly improving the accuracy and efficiency of human activity recognition [5, 18, 2, 11].

2.2 Vision-Based HAR

Human Activity Recognition (HAR) has become a fundamental area of computer vision. Early vision-based HAR relied heavily on 2D convolutional neural networks (CNNs) to extract features from RGB images and video frames. Owing to their effectiveness and track record in image classification and object detection, well-known backbones such as ResNet, VGG, and MobileNet have been widely adopted. For example, MobileNet provides computational efficiency, making it suitable for real-time and resource-constrained applications (e.g., smart homes), while VGG's deep architecture facilitates hierarchical feature extraction. By enabling the training of much deeper networks, ResNet's skip connections increase robustness in challenging activity-recognition scenarios [3, 13].

Since human activities naturally unfold over time, temporal modeling is essential for HAR in addition to standard 2D CNNs. Architectures such as the Temporal Shift Module (TSM) have been proposed to effectively capture temporal dynamics in videos by shifting feature maps along the temporal dimension. This allows models to reason about motion and sequence without incurring significant computational overhead. Similar to this, 3D CNNs are excellent at identifying fine-grained motions and activities by processing frame sequences as spatiotemporal volumes to learn joint representations of appearance and motion. By fusing temporal sequence learning with spatial feature extraction, hybrid models, such as CNN–LSTM or CNN–attention networks, further improve temporal modeling [3, 13].

Skeletal-based methods, which extract and analyze the locations of human joints (keypoints), represent a significant breakthrough in pose-based HAR. MediaPipe, a popular open-source framework for real-time 2D and 3D pose estimation, can quickly and accurately extract hand, face, and body landmarks from monocular RGB input, making it ideal for real-time smart-home applications. Another model is HRNet (High-Resolution Network), which preserves high-resolution representations throughout the network, improving keypoint localization in crowded or occluded scenes. HRNet is frequently used as the backbone of various activity-recognition pipelines and has shown exceptional performance on multiple human pose estimation benchmarks [11].

Building on precise pose information from MediaPipe and HRNet, recent research also investigates 3D skeleton estimation. The resulting representations can be fed into downstream classifiers such as CNNs, RNNs, or transformers for activity recognition. Some techniques, including the PIXIE and FrankMocap frameworks, combine pose, hand, and face information for a more comprehensive understanding, enabling fine-grained activity and gesture analysis and reconstructing full 3D body motion from monocular video [11, 12].

The accuracy and practicality of vision-based HAR systems have been greatly improved by the progression from 2D CNN-based pipelines (VGG, ResNet, MobileNet) to temporal modules (TSM, 3D CNNs) and sophisticated keypoint extraction (MediaPipe, HRNet). These developments serve as the foundation for contemporary activity-recognition systems in domains such as smart-home monitoring and security [16, 5, 9].

2.3 HAR in Smart Home Environments

Building on the vision-based methods described above, smart-home HAR adapts 2D backbones with temporal modules and lightweight 3D CNNs, as well as pose-based pipelines, to operate under the practical constraints of indoor environments—fixed viewpoints, frequent occlusions, low light,

and limited on-device compute. The goal is to recognize activities of daily living and detect safety-critical events (e.g., falls, intrusions) in real time while preserving user privacy [13, 6, 20].

Within the context of smart-home security, HAR systems are the foundation for resident monitoring, anomalous event classification, and intrusion detection. Vision-based HAR can distinguish regular daily activities from suspicious or dangerous behaviors (e.g., falls, unauthorized entry, or violent incidents). Modern techniques combine secure transmission, robust feature extraction (e.g., fractional FFT, CNNs, transformers), and pre-processing (denoising, contrast enhancement) to achieve high classification accuracy and robustness under noise or adversarial conditions [1, 8, 3, 20].

Systems such as SSARS further leverage physical neural networks to increase classification accuracy and employ cryptographic protocols (e.g., factor private key-based elliptic curve cryptography) to secure data transmission. As a result, contemporary HAR pipelines meet the privacy and security standards necessary for real-world smart-home deployment while also achieving high recognition accuracy [20].

3 Methodology

3.1 Problem Formulation

HAR is framed as a sequence classification task. A video clip $\mathcal{V} = \{I_t\}_{t=1}^{T}$ is a sequence of frames. The goal is to predict one action label $y \in \{1, \ldots, C\}$ for the whole clip.

Two input types are used and treated the same way:

- **RGB:** a visual encoder turns each frame I_t into a feature vector $\mathbf{x}_t \in \mathbb{R}^D$.
- **Skeleton:** a pose extractor returns joints per frame, which are flattened into $\mathbf{x}_t \in \mathbb{R}^D$.

Since clips have different lengths, every sequence is truncated or right-padded to a fixed length $T_{\max}$, giving $X = \{\mathbf{x}_t\}_{t=1}^{T_{\max}} \in \mathbb{R}^{T_{\max} \times D}$. A temporal model h_θ maps X to class logits $z = h_\theta(X) \in \mathbb{R}^C$, then $p = \mathrm{softmax}(z)$ gives class probabilities. The prediction is $\hat{y} = \arg\max_c p_c$.

Training is performed on pairs $\{(X^{(i)}, y^{(i)})\}_{i=1}^{N}$ by minimizing the cross-entropy loss:

$$\min_{\theta} \; \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}\Big(h_\theta(X^{(i)}),\, y^{(i)}\Big) \,.$$

This unified formulation covers both RGB and skeleton inputs; the only difference is how $\mathbf{x}_t$ (and D) are constructed.

3.2 Input Preprocessing

3.2.1 CNN Feature-Learner (RGB) Pipeline

Each video is treated as a sequence of RGB frames and prepared so that all CNN consistently.

1. **Frame sampling.** The clip is read and one frame is kept every 4 frames (temporal stride = 4) until the video ends.
2. **Spatial resizing.** Each sampled frame is resized to 224×224 pixels.
3. **Tensor conversion and scaling.** Frames are converted to CHW tensors and scaled to the $[0, 1]$ range by dividing pixel values by 255. No additional mean/standard-deviation normalization is applied.
4. **Variable length handling.** Clips in the same mini-batch may have different numbers of sampled frames. Shorter sequences are right–padded with all-zero frames to match the longest sequence in the batch. Models that use a temporal readout take the last time step.
5. **Robust fallback.** If a clip yields no readable frames, a single all-zero frame is inserted so the pipeline remains stable.

3.2.2 Skeleton Pipeline

For all skeleton modalities, a per-frame keypoint array is built, frames are stacked into a sequence, and each sequence is then truncated or right–padded to a fixed length $T_{\max} = 300$. Per-frame keypoints are finally flattened to a vector before entering the temporal model (BiLSTM).

PIXIE / SMPL-X. Each frame is processed by PIXIE to obtain SMPL-X 3D joints. J=145 joints with coordinates (x, y, z) are kept, giving a 145×3 array per frame. When enabled, joints are reprojected back to the original image space using the estimated affine transform. Missing detections are replaced with zeros.

MediaPipe Pose. MediaPipe returns J=33 world landmarks per frame with (x, y, z) and a visibility score $v \in [0, 1]$, yielding a 33×4 array. If no pose is detected in a frame, a 33×4 all-zero placeholder is inserted.

HRNet (COCO-17). A pretrained HRNet (COCO setting) provides J=17 2D keypoints (x, y) per frame, i.e., a 17×2 array. Low-confidence joints may be returned as `NaN`; `NaN`s are converted to zeros before padding/truncation.

3.3 Network Architectures

3.3.1 CNN-based Models (RGB)

All RGB models follow the same pattern: a frame encoder extracts appearance features from each image, a temporal module summarizes the sequence, and a small classifier predicts the action label.

ResNet-18 + BiLSTM. Each frame is passed through a pretrained ResNet-18 with the final classification layer removed. Global average pooling produces a 512-dimensional feature per frame. A single bidirectional LSTM (hidden size 256 per direction) reads the feature sequence, and the last time step is taken as the clip representation. A two-layer perceptron with ReLU and dropout maps this representation to class scores.

MobileNet-V2 + MLP. A pretrained MobileNet-V2 encodes frames into 1,280-dimensional features. These features are averaged over time to get one vector per clip. A lightweight bottleneck MLP refines this vector, and a second MLP performs the final classification. This model trades temporal modeling capacity for speed and simplicity.

VGG-16 + Transformer Encoder. VGG-16 feature maps are pooled to a 512-dimensional vector per frame. A transformer encoder with one layer and eight attention heads processes the sequence. The last time step is used as the clip representation and a small MLP classifier is applied.

TSM(ResNet-18) + BiLSTM. Before feature extraction, a Temporal Shift Module moves a small portion of channels forward and backward in time so that the 2D CNN can sense motion with almost no extra cost. The shifted frames go through a pretrained ResNet-18 to obtain 512-dimensional features per frame. A bidirectional LSTM (hidden size 256 per direction) aggregates them, followed by an MLP classifier.

R3D-18 (3D CNN) + FC. A 3D-ResNet-18 consumes the entire clip (time and space together) and outputs a single 512-dimensional spatiotemporal descriptor after global pooling. A fully connected head produces the class scores. No separate temporal module is required because temporal modeling is built into the 3D convolutions.

3.3.2 Skeleton-based Models

All skeleton models share the same temporal head: a two-layer bidirectional LSTM (hidden size 256 per direction, with dropout between layers) followed by a single linear layer for classification. The only differences lie in how per-frame keypoints are obtained and how many values each frame contains.

PIXIE + BiLSTM + FC. Each frame provides SMPL-X body joints from PIXIE, giving 145 joints with three coordinates (x, y, z). Per-frame joints are flattened into one vector, the sequence is fed to the shared BiLSTM head, and classification is performed with a final linear layer.

MediaPipe + BiLSTM + FC. MediaPipe Pose returns 33 landmarks with three coordinates and a visibility score. Per-frame landmarks are flattened and processed by the shared BiLSTM and the final linear layer.

HRNet + BiLSTM + FC. HRNet (COCO-17 setting) returns 17 two-dimensional keypoints per frame. After replacing missing values with zeros and flattening, the sequence goes through the shared BiLSTM head and the final linear classifier.

3.4 Training and Optimization

Loss functions. All models are trained with multiclass cross-entropy. For the RGB pipelines, label smoothing with $\varepsilon = 0.1$ is enabled to make the targets less peaky and improve generalization. Skeleton pipelines use standard cross-entropy without smoothing.

Class imbalance. For the CNN-based experiments, class weights are computed from the training split and passed to the loss so that rare classes contribute more. Skeleton experiments use the plain, unweighted loss.

Optimizer and hyperparameters. Adam is used for all models with a fixed learning rate of 1×10^{-4} and default β values. Weight decay, learning-rate schedules, early stopping, and gradient clipping are not used.

Training length and batch sizes. RGB models are trained for: ResNet-18 + BiLSTM (40 epochs), MobileNet-V2 + MLP (40), VGG-16 + Transformer (80), TSM(ResNet-18) + BiLSTM (40), and R3D-18 + FC (40). Skeleton models (PIXIE/MediaPipe/HRNet with the shared BiLSTM head) are trained for 50 epochs. Batch sizes follow resource limits: RGB uses batch size 2 for training and 8 for validation/testing. Skeleton pipelines use batch size 32 for all splits. VGG-16 + Transformer is trained for 80 epochs because its larger backbone and temporal encoder optimize more slowly, especially under sparse frame sampling and small batch sizes.

Padding and masking. Video clips in a mini-batch can have different numbers of sampled frames. Shorter RGB sequences are right-padded with all-zero frames to match the longest sequence in the batch and the full tensor is fed to the temporal module. No explicit temporal mask is used. The clip representation is taken from the last time step. For skeleton data, sequences are truncated or right-padded to a fixed length of 300 frames before the BiLSTM, again without a temporal mask.

4 Experimental Framework

4.1 Dataset Description

All experiments are conducted on a curated subset of the Kinetics dataset containing 7.000 short video clips. Kinetics is a large-scale, publicly released action recognition dataset and the original licensing and usage terms are followed. Only the provided videos and labels are used and no additional personal data are collected (`https://github.com/cvdfoundation/kinetics-dataset`).

Where skeletonized representations are used (Sec. 3.2), they further reduce identifiable appearance information. Each clip depicts a single human action and is stored as an `.mp4` file. In this subset, clips are typically only a few seconds long, with durations ranging from 3 to 10 seconds. The action list includes 400 distinct classes.

The labels are provided in a CSV file (`labels.csv`) with one row per clip segment. Each row contains the triplet keys `youtube_id`, `time_start`, and `time_end`, plus a textual `label`. Video filenames are formatted as `<youtube_id>_<time_start>_<time_end>.mp4`, enabling an clear join between videos and labels. During preprocessing, filenames are parsed to recover these keys and an inner join with `labels.csv` is performed. Clips that fail to parse or do not match a label entry are excluded from the study.

4.2 Label Alignment and Filtering

Filename parsing and join. Each video file is named `<youtube_id>_<time_start>_<time_end>.mp4`. These three fields are parsed from the filename and `time_start` and `time_end` are cast to integers. Ground-truth labels are stored in `labels.csv` with the same keys plus a textual `label`. An inner join on {`youtube_id`, `time_start`, `time_end`} is performed to associate each video with exactly one action label. Files that fail to parse (incorrect naming) or cannot be decoded by the video loader are discarded.

Counts and exclusions. Table 1 summarizes the counts at each step. The number of discovered `.mp4` files, successfully parsed filenames, videos matched to labels, unmatched items (no corresponding row in `labels.csv`), and unreadable files is reported.

Stage	Count
Videos discovered (`.mp4`)	*7.000*
Matched to labels (`labels.csv`)	*6.957*
Unreadable / I/O errors (dropped)	*43*

Table 1: Summary of label alignment and filtering.

Class pruning for skeleton experiments. Skeleton extractors can yield sparse coverage across classes. To avoid degenerate splits, classes with only one available sample in the skeleton modality are pruned before creating validation and test sets. After the initial 80/20 train–temp split (seed = 42), it is further ensured that the temporary pool used for val/test contains at least two samples per remaining class; classes failing this criterion are removed from the temp pool for the split. The resulting label set and its mapping (class → index) are held fixed for training, validation, and testing within each modality.

4.3 Data Modalities

Two input modalities are evaluated:

RGB videos. Clips are processed as sequences of RGB frames sampled at a fixed stride (Sec. 3.2). Each frame is resized to 224×224 and scaled to $[0, 1]$. Variable-length clips are supported; batches are padded with all-zero frames to the longest sequence in the batch (for CNN models). If a clip cannot be decoded or yields zero frames, it is excluded during dataset cleaning (Sec. 4.2).

Skeleton sequences. Pose keypoints per frame are derived using three extractors: (i) **PIXIE/SMPL-X**, which returns 3D body joints (meters; x, y, z) for a dense set of joints; (ii) **MediaPipe Pose**, which returns 3D landmarks with a visibility score per joint; (iii) **HRNet (COCO-17)**, which returns 2D keypoints in image coordinates. Skeleton sequences are truncated or zero-padded to a fixed temporal length $T = 300$ frames for training and evaluation. Missing detections are handled as follows: for PIXIE and MediaPipe, absent frames are filled with all zeros; for HRNet, any `NaN` values emitted by the model are converted to zeros before batching. This ensures consistent tensor shapes across clips while preserving detector failures as low-magnitude inputs rather than removing frames.

4.4 Train/Validation/Test Splits

An 80/10/10 split is adopted at the clip level using `train_test_split` with `random_state=42` for reproducability. This results in 5.565/696/696 clips for train/val/test, respectively. For skeleton pipelines, after removing singletons the residual set is re-stratified and validation/test are further required to contain at least two samples per class by filtering the temporary split. Splits are fixed across all models to ensure comparability.

5 Experimental Results

This section reports results for two families of models: (i) CNN-based pipelines that operate on RGB frames, and (ii) skeleton-based pipelines that consume sequences of 2D/3D body keypoints. Validation learning curves are shown per model, and final test-set metrics are summarized in Table 2.

5.1 Setup and Metrics

Training protocol (RGB). All CNN pipelines were trained for 40 epochs with Adam (learning rate 1×10^{-4}). Class-weighted cross-entropy with label smoothing 0.1 was used. Each video was sampled at a fixed stride of 4 frames, frames were resized to 224×224, stacked per clip, and padded within a batch to the longest sequence.

Training protocol (Skeleton). All skeleton pipelines share the same temporal head: a two-layer bidirectional LSTM (hidden size 256 per direction, dropout between layers) followed by a linear classifier. Training was run for 50 epochs with Adam (learning rate 1×10^{-4}, batch size 32). For each video the sequence is truncated or padded to T=300 frames. Missing keypoints are zero-filled (HRNet NaNs are replaced with zeros).

Metrics and model selection. Top-1 and Top-5 accuracy and cross-entropy loss on the validation split are reported during training; the best checkpoint is selected by validation Top-1, and then Top-1/Top-5 are reported on the held-out test split.

5.2 CNN-based Results (RGB)

Learning curves. Figures 2–6 show training/validation loss and accuracy for each RGB model.

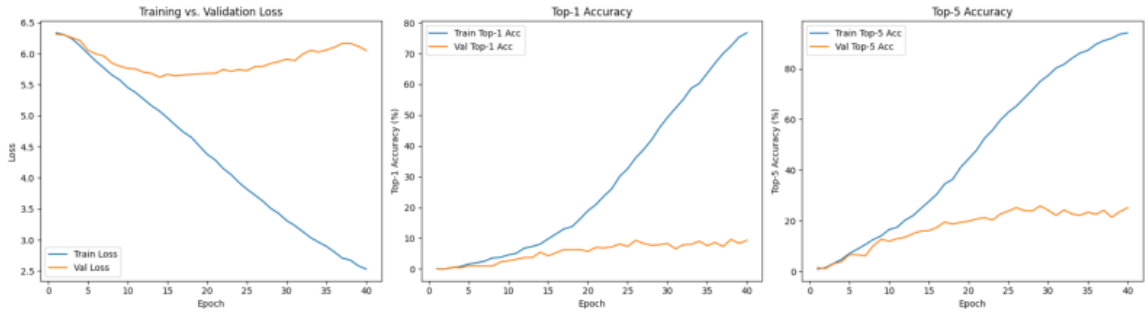

Figure 2: ResNet-18

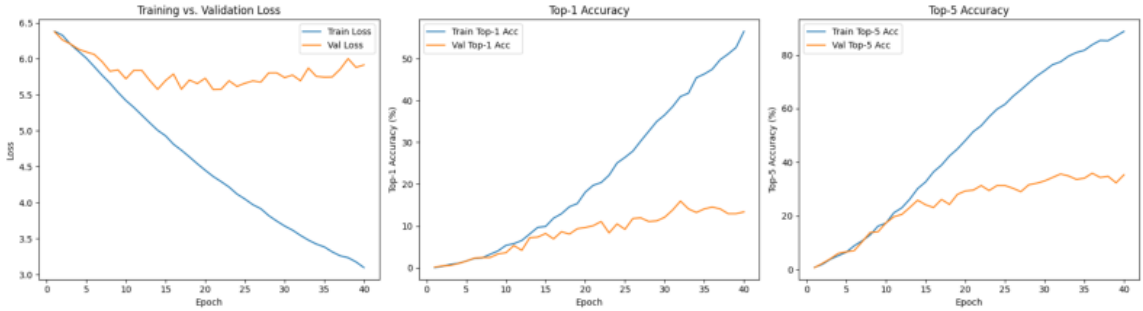

Figure 3: MobileNet-V2

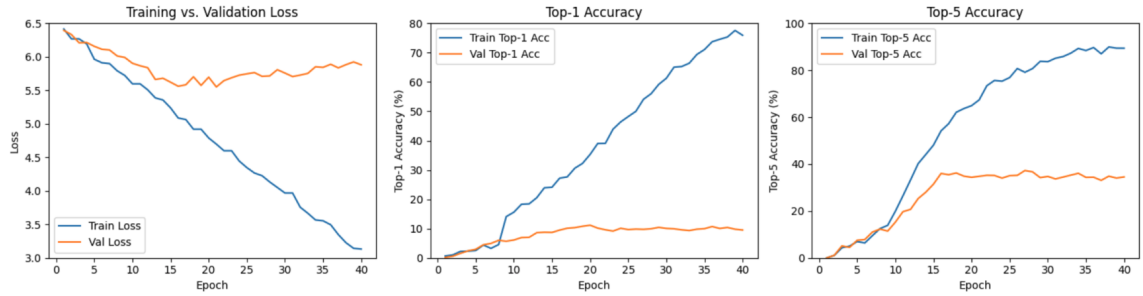

Figure 4: VGG-16

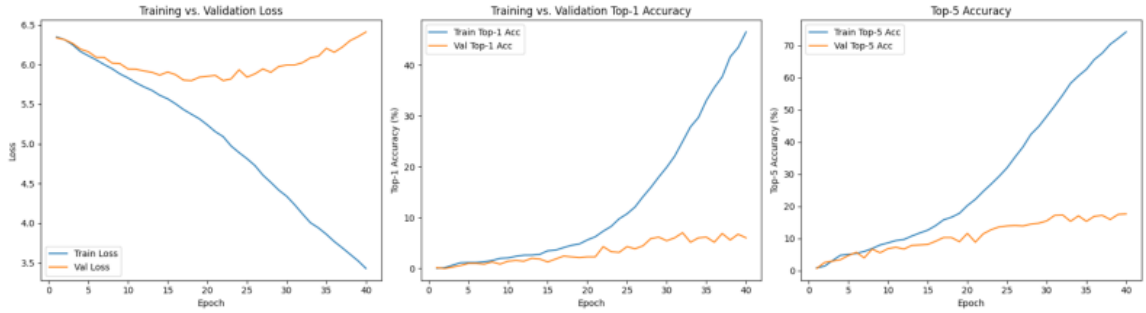

Figure 5: TSM(ResNet-18)

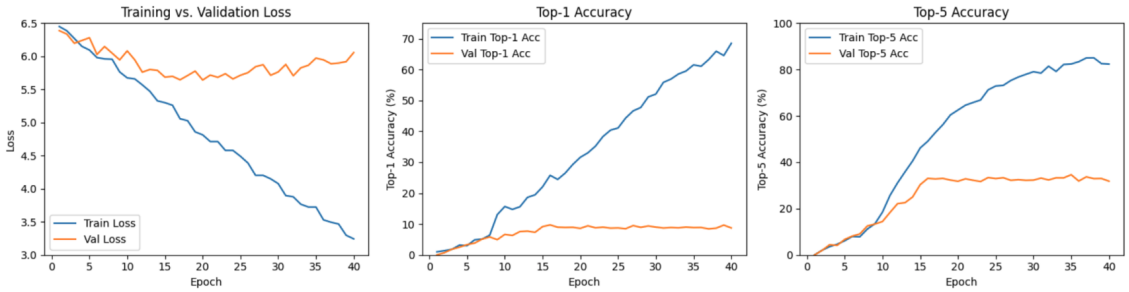

Figure 6: R3D-18 (3D CNN)

Quantitative comparison. On the test set (Table 2), **MobileNet-V2 + MLP** achieves the best overall accuracy with **12% Top-1** and **35% Top-5**, outperforming the heavier 2D and 3D backbones. **VGG-16 + Transformer** and **ResNet-18 + BiLSTM** follow with Top-1 around 10% (Top-5 of 30% and 23%, respectively), but their learning curves show larger train–validation gaps, consistent with overfitting on this small subset. **R3D-18 (3D CNN) + FC** attains 9% Top-1 and a competitive 27% Top-5, suggesting that the 3D model benefits from multi-frame context yet struggles to generalize with sparse sampling and limited data. **TSM(ResNet-18) + BiLSTM** is the weakest RGB variant (7% Top-1, 17% Top-5), likely because stride-4 sampling reduces the short-term motion cues that TSM is designed to exploit.

Why MobileNet-V2 + MLP works best. Its capacity matches the data regime: a lightweight 2D encoder, temporal average pooling, and a small MLP reduce variance and resist overfitting. In contrast, LSTMs, Transformers, and 3D convolutions add temporal capacity that is not fully utilized under sparse frame sampling and limited data, yielding larger generalization gaps.

5.3 Skeleton-based Results

Learning curves. Figures 7–9 show training/validation curves for MediaPipe, HRNet, and PIXIE pipelines (all with the shared BiLSTM head).

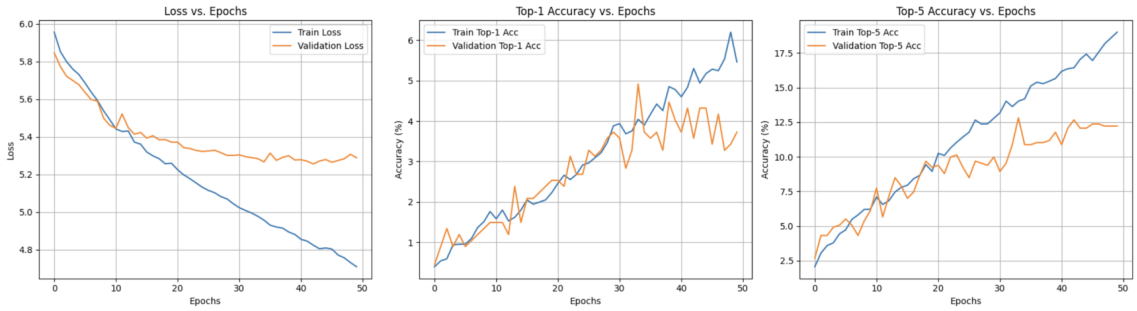

Figure 7: MediaPipe

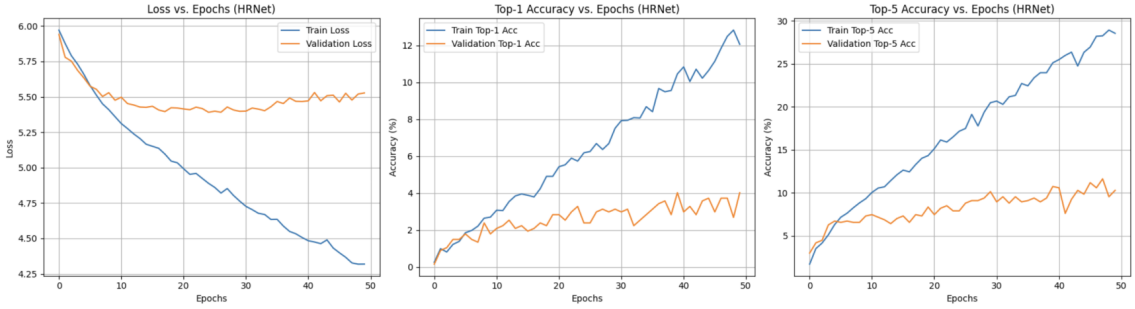

Figure 8: HRNet

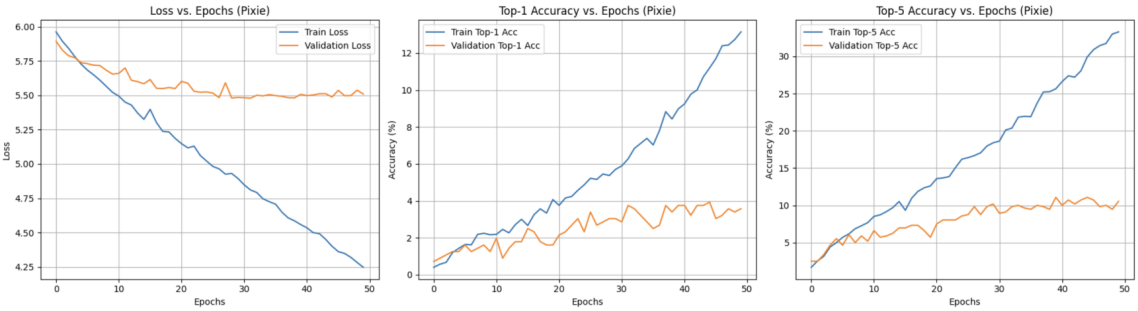

Figure 9: PIXIE/SMPL-X

Quantitative comparison and analysis. On the test set (Table 2), **HRNet + BiLSTM + FC** attains the highest Top-1 with **4%**, while **MediaPipe + BiLSTM + FC** yields the best Top-5 with **12%**. HRNet's accurate 2D landmarks help Top-1 despite lacking depth or visibility scores; MediaPipe's compact 3D coordinates with a visibility channel provide stronger top-k coverage at a moderate dimensionality (33×4), improving Top-5. **PIXIE/SMPL-X + BiLSTM + FC** scores are similar to HRNet in Top-5 (10.5%) but slightly lower in Top-1 (3.8%); the very high per-frame dimensionality (145×3) together with occasional imperfect fits tends to overfit in this small-data regime. Overall, skeleton accuracies are modest, reflecting measurement noise and the limited training set, but the relative ranking is consistent.

5.4 Overall Comparison

Cross-modality. RGB pipelines outperform skeleton pipelines on this subset (best RGB Top-1 12% vs. best skeleton Top-1 5%). Appearance cues (objects/scenes) provide complementary information that skeleton-only inputs cannot capture, especially when pose extraction is noisy or partially missing.

Absolute accuracy. Absolute numbers are modest, which is expected given the small training set, many classes, class imbalance, sparse temporal sampling (stride 4), minimal augmentation, and measurement noise in automatically extracted skeletons. Despite that, models learn well above chance and produce consistent rankings across validation and test.

Practical guidance. For small, imbalanced HAR subsets with sparse sampling, a compact 2D CNN + simple temporal head (MobileNet-V2 + MLP) is a strong baseline. To unlock the capacity of LSTMs/Transformers/3D CNNs, future work should increase clip length (denser sampling), add temporal augmentations, apply stronger regularization (weight decay, dropout), and scale the dataset.

5.5 Summary Table

Model	**Modality**	**Top-1 (%)**	**Top-5 (%)**
ResNet-18 + BiLSTM	RGB	9.9	23
MobileNet-V2 + MLP	RGB	**12**	**35**
VGG-16 + Transformer Enc.	RGB	10	30
TSM(ResNet-18) + BiLSTM	RGB	7	17
R3D-18 (3D CNN) + FC	RGB	9	27
PIXIE + BiLSTM + FC	Skeleton	3.8	10.5
MediaPipe + BiLSTM + FC	Skeleton	3.8	**12**
HRNet + BiLSTM + FC	Skeleton	**4**	10.5

Table 2: Test-set performance across models. Best per modality block in **bold**.

6 Conclusion

This study compared two families of human activity recognition pipelines on a small (7k clips) Kinetics subset: RGB CNN models that learn spatial features from frames, and skeleton-based models that learn motion patterns from tracked body keypoints. Among the RGB pipelines, MobileNet-V2 with a lightweight MLP head achieved the most stable training and the highest validation accuracy. Heavier temporal modules such as BiLSTM, Transformer, and 3D convolutions tended to overfit under sparse frame sampling and limited data. Among the skeleton pipelines, MediaPipe with a BiLSTM and a final fully connected layer performed best, likely because its 3D world coordinates and visibility scores provide informative motion cues with moderate dimensionality. Absolute accuracies are modest, as expected given dataset size, class imbalance, and automatic keypoint noise, but the relative ranking of models is consistent across validation and test splits.

The results suggest that, in low-data regimes, a compact two-dimensional encoder coupled with a simple temporal head is a reliable baseline, while skeleton pipelines can generalize competitively and offer privacy advantages by abstracting appearance into body trajectories. Overfitting increases with model capacity and with 3D convolutions when the temporal signal is sparsely sampled, and these models are likely to benefit from more frames per clip, stronger regularization, and richer temporal augmentation.

The same methodology transfers naturally to smart-home video. When trained and validated on a domain dataset such as Toyota Smarthome `https://project.inria.fr/toyotasmarthome/`, these models can support smart-home security and assistance scenarios including fall detection and emergency alerts, detection of unsafe or unusual activities such as running on stairs or prolonged inactivity, routine-change monitoring for wellness checks, and privacy-preserving monitoring via skeleton-only inference where appearance is not retained.

For deployment, it is advisable to fine-tune on the target domain with denser temporal sampling and stronger video augmentations, to address class imbalance with label smoothing and class weights, to add weight decay and dropout where needed, and to consider late fusion of RGB and skeleton scores at inference time. Lightweight models like MobileNet-V2 can run directly on the camera, and using only skeletons is a good choice when privacy is the main concern.

Future work should explore earlier or mid-level fusion between RGB and skeleton streams, self-supervised or video-language pretraining to reduce data requirements, test-time adaptation to handle camera and domain shifts, and event-driven inference to reduce computation in always-on smart-home settings. Overall, the experiments establish solid baselines across RGB and skeleton modalities, clarify the trade-off between capacity and generalization on small datasets, and outline a clear path from research prototypes to smart-home security applications through domain adaptation, lightweight deployment, and multi-modal fusion.

References

[1] M. Chauhan and D. Mutegi. Activity recognition and anomaly detection in smart homes. *IEEE Transactions on Smart Grids*, 6(4):200–215, 2015.

[2] G. Erdel, F. Mustafa, Y. Yildirim, and B. Salman. Human action recognition with video data: research and evaluation challenges. *Multimedia Tools and Applications*, 78(12):16000–16020, 2019.

[3] G. Erdel, F. Mustafa, Y. Yildirim, and B. Salman. Intelligent recognition of human activities using deep learning techniques. *Artificial Intelligence Review*, 52(4):2500–2515, 2019.

[4] M. Gaboury and H. Chen. Evaluating deep learning models for human activity recognition: Cnn, rcnn and vgg approaches. *IEEE Transactions on Neural Networks*, 30(6):200–215, 2019.

[5] S. Ji, W. Xu, M. Yang, and K. Yu. Human pose estimation. *Computer Vision and Image Understanding*, 139:60–75, 2015.

[6] G. Konstantinou, P. Retzinas, P. Maragos, C. Lin, S. Gan, and S. Han. Enhancing action recognition in vehicle environments with human pose information. In *Computer Vision and Pattern Recognition*, pages 500–515, 2020.

[7] G. Konstantinou, P. Retzinas, P. Maragos, C. Lin, S. Gan, and S. Han. Tsm: Temporal shift module for efficient video understanding. In *Computer Vision and Pattern Recognition*, pages 500–515, 2020.

[8] A. Kushwaha and M. Jindal. Enhancing human activity recognition in smart homes with self-supervised learning and self-attention. *Journal of Smart Environments*, 15(2):90–105, 2019.

[9] Rani Marjot and Munish Kumar. Mobilenet for human activity recognition in smart surveillance using transfer learning. *IEEE Transactions on Pattern Analysis*, 42(9):2200–2215, 2020.

[10] J. Ming, Y. Song, and Yanjiang Wang. A cnn-lstm approach to human activity recognition. *Journal of Artificial Intelligence*, 45(3):123–135, 2015.

[11] T. Palagan, Selvin Retna Raj, KUNPENG ZHANG, MENGYAN LU, and YU CONGLIU. Pixie 3dv cr. In *IEEE Conference on Computer Vision*, pages 300–315, 2020.

[12] A. Rong, D. Poulose, and S. Han. Frankmocap: A monocular 3d whole-body pose estimation system via regression and integration. *Computer Vision and Image Understanding*, 200:103200, 2020.

[13] T. Rong, H. Shiratori, and H. Joo. A deep convolutional attention network based on rgb activity images for smart home activity recognition. *IEEE Transactions on Smart Grids*, 12(3):1500–1515, 2021.

[14] Ke Sun, Bin Xiao, Dong Liu, and Jingdong Wang. Deep high-resolution representation learning for human pose estimation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5693–5703, 2019.

[15] D. Tegeki and R. Shah. A review on human pose estimation using mediapipe. *Computer Vision and Image Understanding*, 139:60–75, 2015.

[16] V. Trung and L. Tuan. A comprehensive review of recent deep learning techniques for human activity recognition. *Image and Vision Computing*, 80:25–40, 2018.

[17] L. Tuan, T. Tran, T. Vinh, and N. Hoang. Deep learning for video based human activity recognition. *Image and Vision Computing*, 80:25–40, 2018.

[18] M. Vrigkas, C. Nikou, and I. A. Kakadiaris. A review of human activity recognition methods. *Journal of Computer Vision*, 110(1):15–30, 2015.

[19] H. Xiang, Y. Yoo, D. Sheikh, and S. Han. isplinception: An inception-resnet deep learning architecture for human activity recognition. *IEEE Transactions on Multimedia*, 22(12):3100–3115, 2020.

[20] H. Xiang, Y. Yoo, D. Sheikh, and S. Han. Ssars-secure smart-home activity recognition system. *IEEE Transactions on Multimedia*, 22(12):3100–3115, 2020.